Triviality of the Tate–Shafarevich Group of the Elliptic Curve 43a1

A curve-specific verification note

2 August 2026

Abstract

Let

$$E/\mathbf{Q}: \qquad y^2 + y = x^3 + x^2,$$

the elliptic curve with Cremona label 43a1. We prove $\mathrm{Sha}(E/\mathbf{Q}) = 0$. The proof identifies an exactly normalized Heegner point over $K = \mathbf{Q}(\sqrt{-7})$, proves that it generates $E(K)$, verifies every residual-image hypothesis in Kolyvagin's odd-primary bound, and uses a full 2-descent for the 2-primary part. All computer calculations used in the proof are exact-arithmetic or algorithmically certified; scripts, retained outputs, and replay commands are linked. The declared trust boundary is the ordinary one for published theorems and the exact PARI/GP and eclib algorithms named below. This is a result about one specified curve. It is not a proof of the Birch and Swinnerton-Dyer conjecture, does not imply the conjecture for any family, and carries no claim of novelty.

1 Scope and trust statement

The purpose of this note is deliberately narrow. Its only main assertion is the following.

Theorem 1. *For the elliptic curve*

$$E/\mathbf{Q} : y^2 + y = x^3 + x^2$$

with Cremona label 43a1*, one has*

$$\mathrm{Sha}(E/\mathbf{Q}) = 0.$$

The theorem is proved under standard theorem/CAS trust. "Theorem trust" means that the cited published results of Kolyvagin, in the precise form quoted by Cha, and the standard results on semistable residual representations, Tate curves, rational torsion, and finite subgroups of $\mathrm{GL}_2(\mathbf{F}_p)$ are accepted. "CAS trust" means that the documented exact arithmetic, full 2-descent, modular-form, modular parametrization, and saturation algorithms in PARI/GP 2.15.4 and eclib/mwrank are accepted. No floating-point recognition, rounded BSD quotient, LMFDB value of Sha, or unproved database image label is a proof input.

This trust statement is not decorative. The exact CM calculation invokes PARI's modular-form and Taniyama-series constructors. The Selmer upper bounds invoke PARI's full 2-descent. The integral Mordell–Weil basis invokes eclib's descent and saturation. The retained artifacts make those calls replayable, but they do not expand the internals of those systems into a kernel-independent formal proof.

No novelty claim is made. The curve is small, its expected arithmetic is in standard databases, and the ingredients are classical. The point here is an explicit proof packet with its hypotheses and trust boundary exposed.

2 Exact arithmetic data

Put $P = (0,0)$. Direct invariant calculation from the displayed equation gives

$$b_2 = 4, \quad b_4 = 0, \quad b_6 = 1, \quad b_8 = 1, \quad c_4 = 16, \quad \Delta = -43, \quad j = -\frac{4096}{43}.$$

Thus the equation is globally minimal, $N_E = 43$, and E is semistable, with nonsplit multiplicative reduction of type I_1 at 43 and good reduction elsewhere. In particular $v_{43}(\Delta_{\min}) = 1$.

Completing the square gives the 2-division polynomial

$$f_2(X) = 4X^3 + 4X^2 + 1.$$

It has no rational root and has discriminant $-16 \cdot 43$. Hence $E(\mathbf{Q})[2] = 0$, and its splitting field has Galois group $S_3 = \mathrm{GL}_2(\mathbf{F}_2)$. Exact counts give

$$\#E(\mathbf{F}_2) = 5, \qquad \#E(\mathbf{F}_3) = 6.$$

Reduction therefore proves that $E(\mathbf{Q})_{\text{tors}} = 0$. These elementary calculations are replayed without external packages by millennium-prize/birch-swinnerton-dyer/43a1/verify_43a1_exact.py; the retained output is millennium-prize/birch-swinnerton-dyer/43a1/output-exact.txt.

3 The normalized Heegner point

Let

$$K = \mathbf{Q}(\sqrt{-7}), \qquad \tau = \frac{-37 + \sqrt{-7}}{86}.$$

The discriminant -7 is fundamental and has class number one. Moreover

$$37^2 + 7 = 4 \cdot 43 \cdot 8,$$

so 43 splits in K, and the maximal-order Heegner hypothesis holds for $X_0(43)$.

Fix the strong Weil parametrization

$$\pi : X_0(43) \longrightarrow E, \qquad \pi(\infty) = O,$$

with pullback of the Neron differential equal to the normalized newform differential. Let y_K be the trace to K of the maximal-order CM point under this parametrization. This convention fixes the base point, sign, and trace normalization used in Kolyvagin's theorem.

Proposition 2. *With these conventions,*

$$\pi(\tau) = P = (0,0), \qquad y_K = P.$$

Proof. The Hilbert class polynomial is $H_{-7}(T) = T + 3375$. The primitive forms $[43, 37, 8]$ and $[1, 37, 344]$, corresponding to τ and 43τ, both reduce to $[1, 1, 2]$. Therefore

$$j(\tau) = j(43\tau) = -3375.$$

Set $S = j(z) + j(43z)$ and $R = j(z)j(43z)$. At the CM point,

$$S = -6750, \qquad R = 11390625. \tag{1}$$

The exact certificate constructs the Laurent series $X(q), Y(q)$ of the optimal parametrization and verifies

$$Y^2 + Y = X^3 + X^2, \qquad \frac{qX'(q)}{2Y(q)+1} = f(q).$$

It also checks modular degree two, Fricke eigenvalue $+1$, and trivial rational torsion. By Manin–Drinfeld [6], the image of the other rational cusp is torsion, hence zero; thus the parametrization is the normalized Fricke quotient, without an unresolved negation or torsion translation. The functions S and R descend to rational functions on E, belonging respectively to $L(43O)$ and $L(44O)$. Their coefficients are recovered over $\mathbf{Q}$ from formal q-series and independently checked through orders 43 and 44. The Riemann–Roch zero bound turns those finite checks into identities, rather than numerical approximations.

Writing

$$S = A_S(X) + B_S(X)Y, \qquad R = A_R(X) + B_R(X)Y,$$

the exact values needed at $X = 0$ are

$$A_S(0) = -6750, \quad B_S(0) = -22750, \quad A_R(0) = 11390625. \tag{2}$$

The certificate takes resultants with the Weierstrass equation and computes the monic gcd of three nonzero projected resultants. That gcd is exactly X. Hence every common point satisfying (1) has $X = 0$. The curve equation then gives $Y = 0$ or -1, while (2) selects $Y = 0$. Thus the CM fiber is the single point P. Since the class number is one, the trace defining y_K has one summand, proving $y_K = P$.

Every operation just described is exact integer or rational arithmetic after the standard PARI constructors are called. The complete executable certificate is millennium-prize/birch-swinnerton-dyer/43a1/verify_dminus7_cm_exact.gp, and its retained successful output is millennium-prize/birch-swinnerton-dyer/43a1/output-dminus7-exact.txt. □

4 The Mordell–Weil index over the quadratic field

The quadratic twist by -7 has global minimal model

$$E^{(-7)} : [0, -1, 1, -16, -106], \qquad y^2 + y = x^3 - x^2 - 16x - 106.$$

Its discriminant and conductor are $-43 \cdot 7^6$ and $43 \cdot 7^2$. Full multiplication-by-two descent gives

$$\dim_{\mathbf{F}_2} \mathrm{Sel}^{(2)}(E/\mathbf{Q}) = 1, \qquad \dim_{\mathbf{F}_2} \mathrm{Sel}^{(2)}(E^{(-7)}/\mathbf{Q}) = 0. \tag{3}$$

Because both rational 2-torsion groups vanish, (3), together with the point P, gives ranks one and zero respectively. The eigenspace decomposition for $\mathrm{Gal}(K/\mathbf{Q})$ therefore gives $\operatorname{rank} E(K) = 1$.

There is no torsion over K. The prime 2 splits and 3 is inert in K. The exact counts

$$\#E(\mathbf{F}_2) = 5, \qquad \#E(\mathbf{F}_9) = (3+1)^2 - a_3^2 = 12, \qquad a_3 = -2,$$

eliminate odd torsion. The irreducible cubic f_2 remains without a root over a quadratic extension, so $E(K)[2] = 0$, which eliminates all 2-primary torsion.

For $Q \in E(K)$, the point $Q - \sigma Q$ is anti-invariant and corresponds to a rational point on $E^{(-7)}$. That twist has rank zero, so $Q - \sigma Q$ is torsion and hence zero. Consequently

$$E(K) = E(\mathbf{Q}). \tag{4}$$

Eclib's full 2-descent and saturation at 1000-bit precision returns $[0:-1:1]=-P$, reports that the basis is already saturated, and certifies unconditionally that it is the full Mordell–Weil basis. Combining this with (4) proves

$$E(K)_{\mathrm{tors}}=0, \qquad E(K)=\mathbf{Z}P. \tag{5}$$

In particular, Proposition 2 and (5) imply

$$[E(K)_{\mathrm{free}}:\mathbf{Z}y_K]=1. \tag{6}$$

The twist descent and point counts are replayed by millennium-prize/birch-swinnerton-dyer/43a1/verify_43a1_K.gp, with output millennium-prize/birch-swinnerton-dyer/43a1/output-K.txt. The full saturation transcript is millennium-prize/birch-swinnerton-dyer/43a1/output-mwrank-saturation.txt. The detailed certificate and its trust accounting are in millennium-prize/birch-swinnerton-dyer/43a1/K-mordell-weil-certificate.md.

5 Residual surjectivity at every prime

Proposition 3. *For every rational prime p,*

$$\overline{\rho}_{E,p}(G_{\mathbf{Q}})=\mathrm{GL}_2(\mathbf{F}_p).$$

Proof. The case $p=2$ follows from the irreducible 2-division cubic with nonsquare discriminant. For odd p, the type I_1 multiplicative reduction at 43 supplies a nonidentity transvection on $E[p]$. This includes $p=43$: the Tate parameter has valuation one, so its class is not a 43rd power and wild inertia supplies the required unipotent. The nonsplit quadratic twist is unramified and does not alter inertia.

If a semistable elliptic curve over $\mathbf{Q}$ has reducible $E[p]$, then it or its p-isogenous curve has a rational point of order p. Mazur's rational-torsion theorem therefore gives irreducibility for $p\geq 11$. For $p=3,5,7$, direct point counts at 2 and 3 give Frobenius discriminants -4 and -8. Their reductions are nonsquares at the indicated primes:

$$-4\equiv 2 \pmod 3, \qquad -8\equiv 2 \pmod 5, \qquad -4\equiv 3 \pmod 7.$$

Thus $E[p]$ is irreducible for every odd p.

For $p\geq 7$, Dickson's classification implies that an irreducible subgroup of $\mathrm{GL}_2(\mathbf{F}_p)$ containing a transvection contains $\mathrm{SL}_2(\mathbf{F}_p)$. The determinant is the surjective cyclotomic character, so the image is all of $\mathrm{GL}_2(\mathbf{F}_p)$. For $p=3,5$, exhaustive exact enumeration handles the exceptional small groups: after fixing the transvection, every matrix with the required Frobenius trace and determinant generates a group of order 48 and 480 respectively, exactly the orders of $\mathrm{GL}_2(\mathbf{F}_3)$ and $\mathrm{GL}_2(\mathbf{F}_5)$.

The point counts, cubic discriminant, and finite matrix enumerations are replayed by millennium-prize/birch-swinnerton-dyer/verify_cycle261_43a1_residual.py. The theorem-level infinite-prime reduction and the characteristic-43 inertia argument are written out in millennium-prize/birch-swinnerton-dyer/cycle-261-43a1-all-prime-residual-surjectivity.md. No Galois-image database is used. □

6 Odd-primary vanishing

We use Kolyvagin's inequality exactly as printed in Cha [1, Theorem 3, p. 155]. In its standing modular-Heegner setting, if p is odd, y_K has infinite order, and $\mathrm{Gal}(\mathbf{Q}(E[p])/\mathbf{Q}) = \mathrm{GL}_2(\mathbf{F}_p)$, then

$$\mathrm{ord}_p \# \mathrm{Sha}(E/K) \leq 2m_p, \tag{7}$$

where m_p is the largest integer for which $y_K \in p^{m_p} E(K)$ modulo p-torsion points, in Cha's wording. The Kolyvagin theorem package also gives the finiteness needed for the order in (7).

The exact wording matters. Theorem 3 has no condition $p \nmid D_K$, no condition $p \nmid N_E$, and no good-reduction condition at p. Therefore it covers both $p = 7$, which ramifies in K, and $p = 43$, where E has bad multiplicative reduction. Those extra local conditions occur in Cha's later irreducibility-only variant, Theorem 21, not in Theorem 3. The published-text audit, including page anchors and stable DOI links, is millennium-prize/birch-swinnerton-dyer/cycle-262-43a1-kolyvagin-theorem3-literature-audit.md.

By (5), (6), and Proposition 2, $y_K = P$ is non-torsion and $m_p = 0$ for every prime. Proposition 3 supplies the full residual-image hypothesis for every odd p. Equation (7) therefore gives

$$\mathrm{Sha}(E/K)[p^\infty] = 0 \qquad \text{for every odd prime } p. \tag{8}$$

Restriction preserves local triviality and defines

$$\mathrm{res} : \mathrm{Sha}(E/\mathbf{Q})[p^\infty] \longrightarrow \mathrm{Sha}(E/K)[p^\infty].$$

Corestriction after restriction is multiplication by two. Hence restriction is injective on odd-primary groups, and (8) yields

$$\mathrm{Sha}(E/\mathbf{Q})[p^\infty] = 0 \qquad \text{for every odd prime } p. \tag{9}$$

The same restriction argument, together with finite generation of $E(K)$, also transfers the qualitative finiteness conclusion to $\mathrm{Sha}(E/\mathbf{Q})$: its restriction kernel lies in the finite group $H^1(\mathrm{Gal}(K/\mathbf{Q}), E(K))$, and its image lies in finite $\mathrm{Sha}(E/K)$.

7 The 2-primary part

The full multiplication-by-two descent over $\mathbf{Q}$ returns one basis element for the everywhere locally soluble 2-covers. Its explicit affine quartic is

$$C: \quad V^2 = U^4 - 2U^2 + 4U + 1. \tag{10}$$

The exact cover map satisfies the Weierstrass equation identically modulo (10). The point $(U, V) = (0, 1)$ maps to $(1, 1) = -3P$, which represents the nonzero Kummer class of P. Therefore

$$\mathrm{Sel}^{(2)}(E/\mathbf{Q}) = \langle \delta(P) \rangle \simeq \mathbf{F}_2. \tag{11}$$

The Kummer sequence is

$$0 \longrightarrow E(\mathbf{Q})/2E(\mathbf{Q}) \longrightarrow \mathrm{Sel}^{(2)}(E/\mathbf{Q}) \longrightarrow \mathrm{Sha}(E/\mathbf{Q})[2] \longrightarrow 0.$$

Since $E(\mathbf{Q}) = \mathbf{Z}P$, both the first and middle terms have dimension one. Thus

$$\mathrm{Sha}(E/\mathbf{Q})[2] = 0. \tag{12}$$

The exact GP checker verifies the complete basis size, literal quartic, cover map, rational point, Kummer class, and 2-division polynomial: millennium-prize/birch-swinnerton-dyer/43a1/verify_43a1_2descent.gp. The certificate explanation is millennium-prize/birch-swinnerton-dyer/43a1/2descent-certificate.md. The upper-bound assertion in (11) is precisely where trust in PARI's documented full 2-descent implementation enters.

The group $\mathrm{Sha}(E/\mathbf{Q})$ is finite by the preceding section. Every nonzero finite 2-primary group contains a nonzero element of order two. Equation (12) therefore implies

$$\mathrm{Sha}(E/\mathbf{Q})[2^\infty] = 0. \tag{13}$$

8 Proof of the theorem

Equation (9) eliminates every odd-primary component of the finite group $\mathrm{Sha}(E/\mathbf{Q})$, and (13) eliminates its 2-primary component. A finite abelian group is the direct sum of its primary components. Hence

$$\mathrm{Sha}(E/\mathbf{Q}) = 0,$$

which proves Theorem 1.

9 Reproduction ledger

Run the following from the repository root. The commands perform exact checks; they do not query LMFDB for the claimed conclusion.

```
python3 millennium-prize/birch-swinnerton-dyer/43a1/verify_43a1_exact.py
gp -fq millennium-prize/birch-swinnerton-dyer/43a1/verify_43a1_2descent.gp
gp -fq millennium-prize/birch-swinnerton-dyer/43a1/verify_dminus7_cm_exact.gp
gp -fq millennium-prize/birch-swinnerton-dyer/43a1/verify_43a1_K.gp
printf '0 1 1 0 0\n' | mwrank -v 2 -p 1000
python3 millennium-prize/birch-swinnerton-dyer/verify_cycle261_43a1_residual.py
bash scripts/build-paper.sh sha-43a1
```

The decisive expected terminal markers are `ALL_EXACT_2DESCENT_ASSERTIONS_PASSED=1`, `FORMAL_Q_CERTIFICATE_PASSED=1`, `ALL_EXACT_K_RANK_TORSION_ASSERTIONS_PASSED=1`, and `ALL_PRIME_RESIDUAL_SURJECTIVITY_CERTIFICATE=PASS`; mwrank must state that the points were already saturated and that the full Mordell–Weil basis was determined unconditionally.

For convenience, the packet index is millennium-prize/birch-swinnerton-dyer/43a1/README.md. Stable external links for the two theorem sources are Cha's article doi:10.1016/j.jnt.2004.08.009 and Kolyvagin's chapter doi:10.1007/BFb0086267. PARI/GP documentation is at https://pari.math.u-bordeaux.fr/dochtml/html-stable/, and eclib source and documentation are at https://github.com/JohnCremona/eclib.

10 What has not been proved

Theorem 1 is curve-specific. It does not prove BSD for 43a1: no equality of the leading Taylor coefficient with the BSD arithmetic expression is asserted here. It does not prove BSD for all elliptic curves, for all rank-one curves, or for any nontrivial family. It does not constitute a Clay Millennium Prize solution or a reduction of that problem. Database entries for analytic rank, regulator, Tamagawa product, and analytic order of Sha are corroboration only and are absent from the logical chain. No publication-priority or novelty claim is made.

References

[1] B. Cha, *Vanishing of some cohomology groups and bounds for the Shafarevich–Tate groups of elliptic curves*, J. Number Theory **111** (2005), 154–178. doi:10.1016/j.jnt.2004.08.009.

[2] V. A. Kolyvagin, *On the structure of Shafarevich–Tate groups*, in Algebraic Geometry (Chicago, 1989), Lecture Notes in Math. 1479, Springer, 1991, 94–121. doi:10.1007/BFb0086267.

[3] B. Mazur, *Modular curves and the Eisenstein ideal*, Publ. Math. IHES **47** (1977), 33–186.

[4] J.-P. Serre, *Proprietes galoisiennes des points d'ordre fini des courbes elliptiques*, Invent. Math. **15** (1972), 259–331.

[5] J. H. Silverman, *Advanced Topics in the Arithmetic of Elliptic Curves*, Graduate Texts in Mathematics 151, Springer, 1994, Chapter V.

[6] Yu. I. Manin, *Parabolic points and zeta functions of modular curves*, Math. USSR-Izv. **6** (1972), 19–64.